

Name _____

CSCI 332, Fall 2026
Quiz 1—Example

1. (2 points) For each symbol below, write its name or meaning in English. The first is filled in for you as an example.

(a) $\mathcal{P}$: power set

(b) $\in$: _____

(c) $\subseteq$: _____

(d) $\cup$: _____

(e) $\cap$: _____

2. (3 points) Circle True or False for each of the following statements.

(a) $A \cap B = \{x : x \in A \text{ and } x \in B\}$: True or False

(b) $5 \in \{n^2 : n \in \mathbb{Z}\}$: True or False

(c) $\mathcal{P}(S)$ is the set of all elements of S : True or False

3. (5 points) Given the following preference lists as input for the hospital/medical student matching problem:

X: A, B, C A: Y, X, Z

Y: B, A, C B: X, Y, Z

Z: A, B, C C: X, Y, Z

(a) Is the matching $M = \{(X, A), (Y, B), (Z, C)\}$ stable? Circle one: Yes or No

(b) How do you know?

(c) Is the matching $M = \{(X, C), (Y, B), (Z, A)\}$ stable? Circle one: Yes or No

(d) How do you know?